

Phase and Dilation Charges in a $GL(2, \mathbb{C})$ Spinor-Density Formalism

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Abstract

In a $GL(2, \mathbb{C})$ -valued chiral BF-like formalism of Plebanski [7] and Robinson [9] type the trace of the connection is a complex one-form, and separating it into real and imaginary parts gives a non-compact dilation potential alongside the compact phase potential usually identified with the Faraday field. A spinor density of weights $(w, \bar{w})$ couples to the two through the sum and the difference of those weights, here called the dilation weight W and the phase charge Δ . Three results follow. Complex conjugation reverses Δ and preserves W , so a sesquilinear pairing is automatically phase-invariant while carrying dilation weight $2W$; with a weight-neutral volume element there is therefore no invariant sesquilinear kinetic term for any field of non-zero dilation weight. The normalisation that ties the $GL(2, \mathbb{C})$ frame bundle to $SO^+(1, 3)$ is exactly the condition that the co-frame carry $W = \Delta = 0$, which makes the volume element weight-neutral, so that condition applies to the theory as usually written. Invariance of the fermionic Lagrangian then delivers $W(\lambda_A) = 0$ directly, which shows that the restriction $GL(2, \mathbb{C}) \rightarrow SL(2, \mathbb{C}) \otimes U(1)$ imposed to obtain the charged Weyl equations is a sufficient condition standing in for a representation-theoretic decoupling that the action already supplies. Together these give a small no-go: within this action the $\mathbb{R}^+$ factor acts trivially on matter and is pure gauge. Three exits are identified and their costs stated, one of which makes any dilation-charged scalar a compensator rather than a matter field. A closing section notes that the separation of a compact charge-like channel from a non-compact scale-like one identifies a mathematically natural neutral sector, and that whether that sector can carry stress-energy and dynamics is the physical question the no-go leaves open.

Keywords: chiral BF, spinor density, Weyl dilation, non-metricity, compensator fields

1 Scope

The algebraic content of this note is the splitting

$$\mathfrak{gl}(2, \mathbb{C}) = \mathfrak{sl}(2, \mathbb{C}) \oplus (\mathbb{R} \oplus i\mathbb{R}), \quad (1)$$

together with the observation that matter representations can distinguish the two real trace channels, one compact and charge-like, the other non-compact and scale-like. A sector neutral under the first and charged under the second is a natural object to ask about, and what follows establishes what the formalism permits such a sector to be. The setting is the chiral BF-like formalism of Plebanski [7], its $\mathfrak{gl}(2, \mathbb{C})$ -valued extension by Robinson [9, 10], and the spinor-density machinery of Plebanski's unpublished notes [8], collected and developed in [1, 2, 3, 4, 5]. The formalism is chiral in the sense that the simplicity conditions on the

two-forms are imposed off shell; the non-chiral alternative, in which they arise dynamically, is that of Smolin and Speziale [20].

Sections 2 and 3 fix the group and the weight conventions and derive the transformation law that everything else rests on. Section 4 records the neutrality of the geometric skeleton and the non-metricity carried by the dilation. Section 5 is the invariance ledger and contains the main results. Section 6 lists the three ways out of the resulting no-go and prices each. Section 7 states the one identification that would tie the construction back to the multiplier sector and that remains unproved. Section 8 says what the separation gives and what it does not.

2 Group structure

The spin structure group for spinor dyads is $\mathrm{SL}(2, \mathbb{C})$, a six parameter subgroup of the eight parameter real Lie group $\mathrm{GL}(2, \mathbb{C})$, which in the Clifford presentation is the group $C\ell_3^*$ of rotors [21, 3]. Any $\xi \in \mathrm{GL}(2, \mathbb{C})$ admits

$$\xi^A{}_B = f L^A{}_B, \quad f^2 = \det(\xi^A{}_B), \quad L \in \mathrm{SL}(2, \mathbb{C}). \quad (2)$$

Remark 2.1 (Not a direct product). Equation (2) requires a choice of square root, and (f, L) and $(-f, -L)$ give the same ξ , so that $\mathrm{GL}(2, \mathbb{C}) \cong (\mathrm{SL}(2, \mathbb{C}) \times \mathbb{C}^*)/\mathbb{Z}_2$. All statements below are made about the algebra and about local sections and are insensitive to this, but the distinction is not optional for any attempt to read the holonomy of the dilation potential globally.

Writing $\gamma^A{}_B$ for the $\mathfrak{gl}(2, \mathbb{C})$ -valued connection one-form,

$$\gamma^A{}_B = \Gamma^A{}_B + \frac{1}{2} \delta^A{}_B \gamma, \quad \gamma := \gamma^C{}_C, \quad \Gamma^S{}_S = 0, \quad (3)$$

and under a change of dyad

$$\gamma^{\tilde{A}}{}_{\tilde{B}} = \xi^{\tilde{A}}{}_A (\xi^{-1})^{\tilde{B}}{}_B \gamma^A{}_B + \xi^{\tilde{A}}{}_S d(\xi^{-1})^{\tilde{B}}{}_S, \quad \tilde{\gamma} = \gamma - d \ln \xi, \quad (4)$$

an abelian gauge transformation of the trace with complex parameter $\ln \xi$. Splitting the trace,

$$\gamma = \mathfrak{a} + i\mathfrak{b}, \quad \mathfrak{a} \mapsto \mathfrak{a} - d \ln |\xi|, \quad \mathfrak{b} \mapsto \mathfrak{b} - d \arg \xi, \quad (5)$$

so that $\mathfrak{b}$ is a compact $\mathrm{U}(1)$ potential and $\mathfrak{a}$ a non-compact $\mathbb{R}^+$ potential.

3 Weights, and how they transform

A generic spinor density, in the sense of Plebanski's unpublished notes [8] as collected in [3, App. 1], transforms as

$$\psi^{\tilde{A}\tilde{B}'}{}_{\tilde{C}\tilde{D}'} = (f)^{2w} (\bar{f})^{2\bar{w}} (\xi^{-1})^{\tilde{C}}{}_C \xi^{\tilde{A}}{}_A (\xi^{-1})^{\tilde{D}'}{}_{D'} \xi^{\tilde{B}'}{}_{B'} \psi^{AB'}{}_{CD'}, \quad (6)$$

with $(w, \bar{w})$ the weight and anti-weight and $\mathrm{SL}(2, \mathbb{C})$ spinors transforming at vanishing weight. Because each index transformation in (6) itself carries a factor of f or $\bar{f}$ through (2), the total power of f attaching to a density with k, p upper and lower unprimed and l, q upper and lower primed indices is $2P$ and of $\bar{f}$ is $2Q$, where

$$P := w + \frac{1}{2}(k - p), \quad Q := \bar{w} + \frac{1}{2}(l - q). \quad (7)$$

Definition 3.1 (Dilation weight and phase charge). Set $W(\psi) := P + Q$ and $\Delta(\psi) := P - Q$.

We call Δ the *phase charge*. Under Robinson's identification [9] of the imaginary part of the trace connection with the Faraday potential, together with the normalisation $\gamma_\mu = -\frac{ie_0}{2\hbar} A_\mu$ of [3], it becomes electric charge in units of $e_0/2\hbar$; that identification is an input from the earlier literature [9, 1], and nothing below depends on it.

Lemma 3.2 (Transformation law). Writing $f = \varrho e^{i\alpha}$ with $\varrho = |f|$,

$$\psi \longmapsto \varrho^{2W(\psi)} e^{2i\alpha\Delta(\psi)} \psi \quad (\text{times the } \text{SL}(2, \mathbb{C}) \text{ action}), \quad (8)$$

and the exterior covariant derivative associated to γ reads

$$\gamma D_\mu \psi = \left(\Gamma \nabla_\mu + W(\psi) \mathbf{a}_\mu + i \Delta(\psi) \mathbf{b}_\mu \right) \psi. \quad (9)$$

Both W and Δ are additive under tensor and wedge products, and both are unchanged by raising or lowering indices with ϵ_{AB} and its relatives, each of which carries $P = Q = 0$.

Proof. The first statement is $f^{2P} \bar{f}^{2Q} = \varrho^{2(P+Q)} e^{2i\alpha(P-Q)}$; the second is (6) differentiated with (5) substituted; additivity is immediate from (6). For ϵ_{AB} , $(w, \bar{w}) = (1, 0)$ and $(k, p; l, q) = (0, 2; 0, 0)$, so $P = 1 - 1 = 0$ and $Q = 0$, and similarly for ϵ^{AB} , $\epsilon_{A'B'}$, $\epsilon^{A'B'}$. $\square$

Lemma 3.3 (Conjugation). For any density, $W(\bar{\psi}) = W(\psi)$ and $\Delta(\bar{\psi}) = -\Delta(\psi)$.

Proof. Complex conjugation exchanges primed and unprimed structure, so it exchanges P and Q ; their sum is invariant and their difference changes sign. $\square$

Lemma 3.3 is the structural asymmetry between the two channels, and it does the work in Section 5. A compact phase is reversed by conjugation, so a sesquilinear pairing is automatically neutral under it; a real scale weight is not reversed, so a sesquilinear pairing doubles it.

4 The skeleton, and the non-metricity

Proposition 4.1. $W(\epsilon_{AB}) = \Delta(\epsilon_{AB}) = 0$; and $\Delta(\theta^{AA'}) = 0$ for any hermitian co-frame, independently of the weight assigned to it.

Proof. The first is Lemma 3.2. For the second, hermiticity requires $w' = \bar{w}$, and $\theta^{AA'}$ carries one upper unprimed and one upper primed index, so $P = Q$ and their difference vanishes. $\square$

Neutrality of the co-frame under the phase is necessary for the phase to admit an electromagnetic reading, since the object from which the metric is derived must not be charged; it is not sufficient, and the sufficiency conditions are those listed after Definition 3.1.

The real linear connection built from the soldering form and the $\text{GL}(2, \mathbb{C})$ connection is $\gamma^a_b = \omega^a_b + 2\delta^a_b \mathbf{a}$, with ω metric. Then

$$\begin{aligned} \Gamma \nabla_\mu g_{ab} &= \partial_\mu g_{ab} - \gamma^c_{a\mu} g_{cb} - \gamma^c_{b\mu} g_{ac} \\ &= (\omega \nabla_\mu g_{ab}) - 2\mathbf{a}_\mu g_{ab} - 2\mathbf{a}_\mu g_{ab} = -4\mathbf{a}_\mu g_{ab}, \end{aligned} \quad (10)$$

the two equal contributions coming from the two indices of g_{ab} . The non-metricity is thus pure trace with coefficient four, which is the geometry of Weyl [16] recovered from the spinor-density side and later revived by Dirac [17]; within the wider metric-affine setting [18] the metric-compatible reduction used in this formalism is exactly $\mathbf{a} = 0$, and Section 5 supplies a reason for it beyond convenience.

5 The invariance ledger

Invariance of the action requires each term of the Lagrangian four-form to carry $W = \Delta = 0$, the coordinate differentials being weightless. What follows uses only Lemma 3.2, additivity, and the stated conventions.

Proposition 5.1 (The morphism fixes the co-frame weight). The condition $(\xi)^{2w+1}(\bar{\xi})^{2\bar{w}+1} = 1$ of [3, App. 2], which is what makes the induced map $\text{GL}(2, \mathbb{C}) \rightarrow \text{SO}^+(1, 3)$ land in the proper orthochronous group and which fixes $w = \bar{w} = -\frac{1}{2}$ for $\theta^{AA'}$, is precisely

$$P(\theta^{AA'}) = Q(\theta^{AA'}) = 0, \quad \text{hence} \quad W(\theta) = \Delta(\theta) = 0. \quad (11)$$

Consequently $W(g_{\mu\nu}) = W(g^{\mu\nu}) = W(\sqrt{g}) = 0$, and the two-forms $\Sigma^A_B = \frac{1}{2}\theta^A_{A'} \wedge \theta^{BA'}$ inherit $W(S^A_B) = 0$, in agreement with the tabulated value.

Proof. With $k = l = 1$ and $p = q = 0$ the total powers of f and $\bar{f}$ attaching to $\theta^{AA'}$ are $2w + 1$ and $2\bar{w} + 1$, so the stated condition is $P = Q = 0$; the rest is additivity together with Lemma 3.2, index position being immaterial. $\square$

Proposition 5.2 (No invariant sesquilinear kinetic term). Let ψ be any density and consider a term of the schematic sesquilinear form $\mathcal{M} \bar{\psi} \psi$, with $\mathcal{M}$ built from the co-frame, the S -forms and the ϵ 's. Then $\Delta(\mathcal{M} \bar{\psi} \psi) = \Delta(\mathcal{M})$ and $W(\mathcal{M} \bar{\psi} \psi) = W(\mathcal{M}) + 2W(\psi)$. Under Proposition 5.1, $W(\mathcal{M}) = 0$, so invariance forces

$$W(\psi) = 0. \quad (12)$$

Proof. Lemma 3.3 gives $\Delta(\bar{\psi} \psi) = 0$ and $W(\bar{\psi} \psi) = 2W(\psi)$; Proposition 5.1 gives $W(\mathcal{M}) = 0$; additivity does the rest. $\square$

Corollary 5.3 (The scalar sector). For the first-order scalar Lagrangian of [2, 3], in which $\sqrt{g}$ and $g^{\mu\nu}$ are the Urbantke functions [14] of the S -forms,

$$L_H = \sqrt{g} \left\{ \bar{\pi}^\mu (\gamma D_\mu \varphi) - \bar{\pi}^\mu \pi^\nu g_{\mu\nu} + \pi^\mu (\gamma D_\mu \bar{\varphi}) \right\}, \quad (13)$$

the first and third terms require $W(\pi) = -W(\varphi)$, while the second requires $W(\pi) = 0$ by Proposition 5.2 applied to π itself. Hence $W(\varphi) = 0$: the scalar density cannot carry dilation weight.

Corollary 5.4 (The fermionic sector). In the chiral matter Lagrangian of [11, 12, 13] in its $\text{GL}(2, \mathbb{C})$ form [10, 3],

$$\begin{aligned} L_{S\rho} = & -2iL_S + \left(S^B_A + \frac{1}{2}\delta^B_A S^E_E \right) \wedge \rho^A \wedge \gamma D \lambda_B + \tau_A^B{}_C \wedge S^A_B \wedge \rho^C \\ & + \frac{3}{32} \lambda_C \lambda^C \rho_A \wedge \rho^B \wedge S^A_B, \end{aligned} \quad (14)$$

with $\rho^C = \theta^{CC'} \tilde{\lambda}_{C'}$ on shell and $W(S^A_B) = W(\theta) = 0$, invariance of the second term gives $W(\rho) + W(\lambda) = 0$, and $W(\rho) = W(\lambda)$ by Lemma 3.3. Hence $W(\lambda_A) = 0$.

Remark 5.5 (Weight balance replaces group restriction). The charged Weyl equations, $D_B{}^{A'} \bar{\lambda}_{A'} = 0$ and $\gamma D^C{}_{D'} \lambda_C = 0$ require that $\mathfrak{a}$ drop out of $\gamma D \lambda$, and it is inferred from this in [11, 1, 3] that the structure group must be reduced, $\text{GL}(2, \mathbb{C}) \rightarrow \text{SL}(2, \mathbb{C}) \otimes \text{U}(1)$, so that $\mathfrak{a} \equiv 0$ for every field at once. By (9) what is actually needed is $W(\lambda_A) = 0$, which Corollary 5.4 supplies from invariance of the fermionic Lagrangian alone. A universal geometrical prohibition is thereby replaced with a representation-theoretic decoupling condition, and the condition is derived rather than assumed.

This concerns an elementary two-component field in this action. It establishes nothing about the effective dilation weights of composite matter, of the mass terms generated after

electroweak symmetry breaking, or of radiative corrections to atomic transition frequencies, and until those are in hand the second-clock objection to Weyl's theory is relocated rather than answered.

Proposition 5.6 (Small no-go). In the action $L_S + 2iL_H + L_{S\rho}$ with the morphism normalisation of Proposition 5.1 and sesquilinear matter pairings, every matter field carries $W = 0$. The dilation potential $\mathfrak{a}$ then acts trivially on all matter, its contribution to γD vanishes identically, and the $\mathbb{R}^+$ factor is pure gauge.

The result explains, rather than merely records, why metric compatibility is imposed in this formalism: with a weight-neutral volume element and sesquilinear matter there is nothing for the dilation to couple to, so setting $Q_{ab} = 0$ discards nothing.

Remark 5.7 (On positivity). The two quadratic terms generated by (9) carry the same sign. With $\gamma D_\mu \varphi = (\nabla_\mu + i\Delta \mathfrak{b}_\mu)\varphi$ the conjugate is $(\nabla_\mu - i\Delta \mathfrak{b}_\mu)\bar{\varphi}$ and the cross term carries $(-i\Delta)(i\Delta) = +\Delta^2$, while in the dilation channel $\bar{\varphi}$ carries the same W by Lemma 3.3 and the coefficient is $+W^2$. No comparison of the two couplings inside a single derivative settles positivity or unitarity; those are questions about the kinetic bilinear and the reality structure of the complete action, and under any of the exits of Section 6 they must be asked afresh, since each exit changes that structure.

6 Reflections

Proposition 5.6 depends on three inputs, and each may be relaxed at a stated cost.

1. *Bilinear rather than sesquilinear pairing.* Lemma 3.3 is the obstruction and it concerns $\bar{\psi}\psi$. A bilinear pairing $\chi\psi$ with $W(\chi) = -W(\psi)$ is invariant, where χ is a partner in the dual weight representation rather than the complex conjugate of ψ . The cost is that the reality structure of the matter sector must be rebuilt, since the hermiticity conditions of Appendix A no longer relate χ to ψ , and whether a positive kinetic bilinear survives that rebuild is the question of Remark 5.7. The asymmetric pairings available once the composition algebra underlying the dyad is enlarged are the subject of [5], which is the natural place for this exit to be pursued.
2. *A weighted volume element.* Relaxing Proposition 5.1, so that θ carries $P = Q = \tau \neq 0$, gives $W(\sqrt{g}g^{\mu\nu}) = 4\tau$ and permits invariance of (13) at $\tau = -W(\varphi)/2$. The cost is that the induced map onto $SO^+(1, 3)$ becomes conformal rather than isometric, so the metric is determined only up to scale; and since τ belongs to the geometry rather than to any matter species, the dilation weight of every sesquilinearly coupled field is then fixed to the single value -2τ . A universal dilation charge is a strong and immediately testable consequence, and it reinstates the second-clock objection for ordinary matter in full.
3. *A compensator.* Allowing the Lagrangian to carry non-zero total weight and integrating it against a volume form built from the scalar itself is the standard conformal device [17, 19], and it works. Its cost is that φ is then a compensator whose function is to absorb the weight, and a compensator is not a matter sector; the gauge-invariant content is the combination $\mathfrak{a} - d\varsigma$ with $\varphi \sim e^{W\varsigma}$.

Only the third has a familiar name, and its Stückelberg reading needs care. If the amplitude of φ is dynamically stabilised at a non-zero value by a potential, or imposed as a non-zero background, the gauge-invariant combination $\mathfrak{a} - d\varsigma$ may acquire a Proca-type term; whether it does is a statement about a potential that this action does not contain. If the trace sector acquires a conventional vector kinetic term, that structure would ordinarily describe three

vector polarisations, and whether the BF constraint structure permits that count is open, since the multiplier Ξ carries constraints whose rank changes when the trace is retained. If the symmetry-breaking scale tends to zero the Stückelberg mass term disappears, and the dilation connection is removable by a gauge choice only where its curvature vanishes; otherwise its surviving modes depend on the kinetic structure inherited from L_S , which has not been extracted.

7 The trace of the multiplier

The multiplier spinor with $\Xi_{ABCD} = \Xi_{CDAB}$ of [9], whose decomposition is given in [3, App. 5], reads

$$\Xi_{ABCD} = \Psi_{ABCD} + 2\epsilon_{B(C}\epsilon_{A|D)}\lambda + \epsilon_{AB}\phi_{CD} + \phi_{AB}\epsilon_{CD} + 2\epsilon_{AB}\epsilon_{CD}k, \quad \lambda := \frac{1}{6}\Xi^{HE}{}_{EH}, \quad (15)$$

and the received treatment [9, 3] sets $k = -\frac{1}{2}$ and then imposes $\Xi^A{}_A{}^B{}_B = \Xi^{AB}{}_{BA}$, which gives $\lambda = 0$.

Open problem 7.1. Determine whether λ is related on shell to the homothetic curvature $\mathfrak{d}\mathfrak{a}$, and whether the condition $\Xi^A{}_A{}^B{}_B = \Xi^{AB}{}_{BA}$ is equivalent to $\mathfrak{a} = 0$.

A proportionality $\lambda \propto *(\mathfrak{d}\mathfrak{a} \wedge \Sigma)$ is suggested by matching component counts and weights and by the fact that the two conditions enter the derivation at the same point. Distinct quantities frequently share a representation and a component count, so no identification is made here; settling it requires the full variation of the $\mathrm{GL}(2, \mathbb{C})$ Lagrangian with respect to $S^A{}_B$ with the trace retained, without linearising and without setting the torsion to zero. The problem gains interest from Proposition 5.6: if λ does encode $\mathfrak{d}\mathfrak{a}$, then the condition that sets $\lambda = 0$ and the weight bookkeeping above are two routes to the same metric compatibility.

8 What the separation gives

The separation of a compact charge-like trace channel from a non-compact scale-like one identifies a mathematically natural neutral sector: a field with $\Delta = 0$ and $W \neq 0$ would be electrically neutral by representation rather than by a tuned coupling, and would couple to a potential sitting in the connection alongside the Faraday potential. Neutral gravitating matter is what the cosmological evidence has been asking for, and the sector is a reasonable thing to have looked for.

What this note establishes is that it is not available for free. Proposition 5.6 says that within the action as it stands the neutral sector is empty, and Section 6 says what must be surrendered to populate it: the sesquilinear reality structure, or the isometric normalisation of the frame together with the non-universality of the dilation charge, or the status of the scalar as matter rather than compensator. Whether any of the three yields a sector with viable stress-energy and dynamics is the physical question left open, and it is well posed. It would be answered in this order: the kinetic bilinear and its positivity, the constraint algebra and degree-of-freedom count with the trace retained, the stress-energy contributed to the physical metric, and only then a weak-field limit. Nothing before that point licenses a comparison with observation, and a mechanism under which every inconvenient measurement involves the wrong kind of matter while every desired astronomical effect remains available through gravity would deserve suspicion from its author before anyone else.

A Spinor-density conventions

The conventions here follow [8] as set out in [3, App. 1–2] and [2]. Transition functions of the $\text{GL}(2, \mathbb{C})$ bundle over M are complex non-singular 2×2 matrices containing eight real parameters, with $f^2 := \det(\xi^A_B)$ and $\bar{f}^2 := \det(\xi^{A'}_{B'})$, and generic densities transform by (6). The Levi-Civita symbol densities carry

	w	$\bar{w}$		w	$\bar{w}$
ϵ_{AB}	1	0	$\epsilon_{A'B'}$	0	1
ϵ^{AB}	-1	0	$\epsilon^{A'B'}$	0	-1

all four having $P = Q = 0$, and hermiticity of an object holds in all spinorial frames only if $w' = \bar{w}$ and $\bar{w}' = w$. The anti-self-dual Faraday two-form is $f_{ab} \leftrightarrow \phi_{AB}\epsilon_{A'B'} + \bar{\phi}_{A'B'}\epsilon_{AB}$ with $\phi_{AB} = \phi_{(AB)}$, which is the decomposition of the field strength into its $(1, 0)$ and $(0, 1)$ parts, and a totally symmetric spinor of rank $2|s|$ represents a massless field of helicity s , the sign fixed by whether the indices are primed or unprimed.

B Metrical gauge to spin group morphism

Demanding $\sigma^{\tilde{A}\tilde{A}'} = \sigma^{\tilde{A}\tilde{A}'}_a \theta^a$ with $\theta^a = T^{-1}{}^a_{\tilde{a}} \theta^{\tilde{a}}$ and $(T_{\tilde{a}}^a) \in \text{SO}^+(1, 3)$ gives

$$T^{\tilde{a}}_a = -\frac{1}{2} \xi^w \bar{\xi}^{\bar{w}} \sigma^{\tilde{a}}_{\tilde{A}\tilde{A}'} \xi^{\tilde{A}}_A \xi^{\tilde{A}'}_{A'} \sigma^{AA'}_a, \quad (16)$$

which holds for arbitrary complex ξ if $(\xi)^{2w+1}(\bar{\xi})^{2\bar{w}+1} = 1$, that is if $P = Q = 0$ as in Proposition 5.1, fixing $w = \bar{w} = -\frac{1}{2}$ for $\theta^{AA'}$.

Two conventions for the co-frame weights appear in [1, 3, 4], one assigning $w = \bar{w} = -\frac{1}{2}$ to $\theta^{AA'}$ and one whose sign changes with index position, giving $P = -1$, $P = 0$ and $P = +1$ for $\theta_{AA'}$, $\theta^A_{A'}$ and $\theta^{AA'}$ respectively. The second cannot be maintained, since the ϵ 's used to move the indices carry $P = Q = 0$ by Lemma 3.2 and index position therefore cannot alter P . The tabulated S -form weights are consistent on this point, all four yielding $P = Q = 0$:

	w	$\bar{w}$	P
S^{AB}	-1	0	$-1 + 1 = 0$
S^A_B	0	0	0
S_{AB}	1	0	$1 - 1 = 0$
$S^{A'B'}$	0	-1	0

so the first convention is adopted throughout, and it agrees with the S -forms through $\Sigma^A_B = \frac{1}{2} \theta^A_{A'} \wedge \theta^{BA'}$.

The finite content of this appendix, together with the index formula $P = w + \frac{1}{2}(k - p)$, $Q = \bar{w} + \frac{1}{2}(l - q)$ of Section 3, the epsilon and S -form tables, and the termwise invariance checks of Section 5, has been transcribed into a python code program that recomputes every charge in exact rational arithmetic and reports the results, including the failure of the second co-frame convention above. The program takes the index formula, the tabulated source weights, the exchange of primed and unprimed data under conjugation and the additivity of the charges as inputs; it verifies the consequences drawn from them here and establishes nothing about whether those inputs describe the geometric objects correctly.

It is available at https://colab.research.google.com/drive/1cwTzzNfuhrbA581KCXhCxgG_LK0mS4bT?usp=sharing.

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